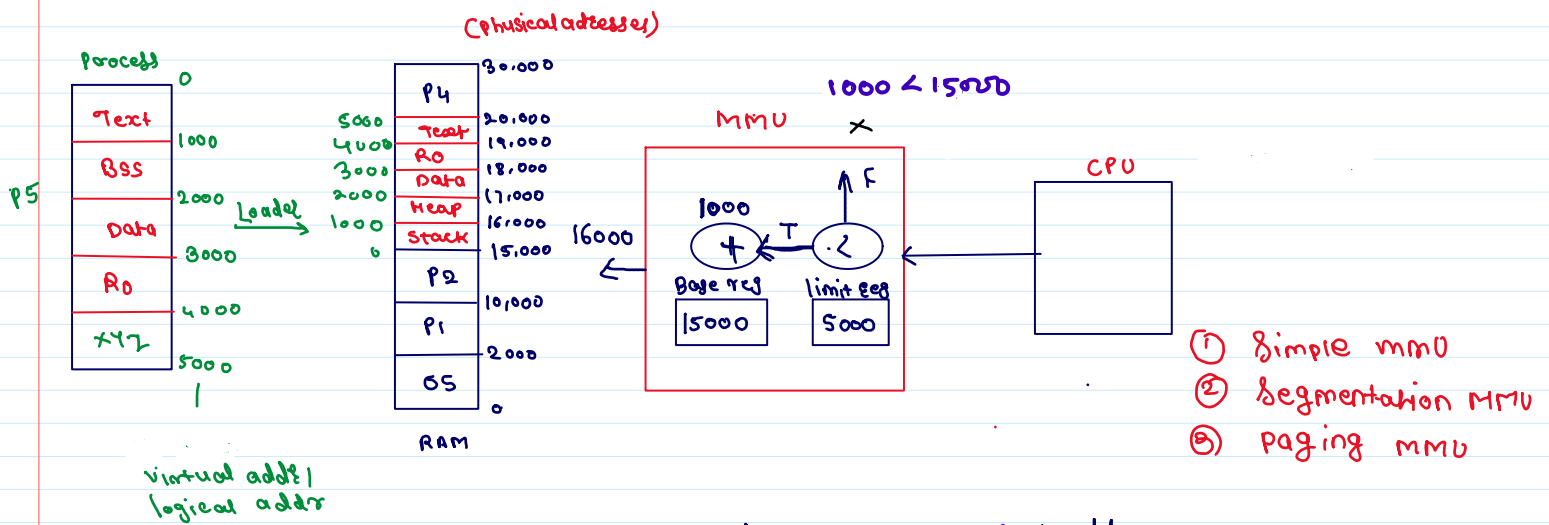


Memory management



- ① Simple MMU
- ② Segmentation MMU
- ③ Paging MMU

MMU - Convert logical ad into physical addr

① memory allocation techniques

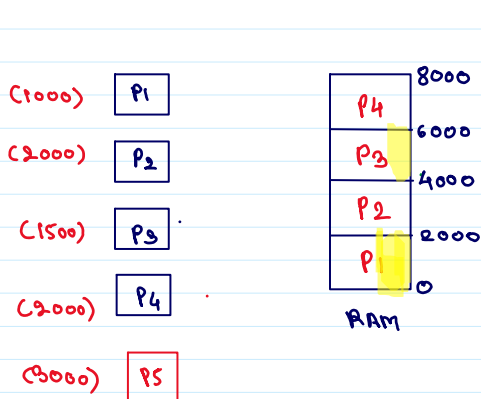
① Contiguous memory allocation

- ① Fixed size partitioning
- ② Variable size partitioning

② non contiguous memory allocation

- ① Segmentation
- ② Paging.

① Fixed size partitioning



2K

P5

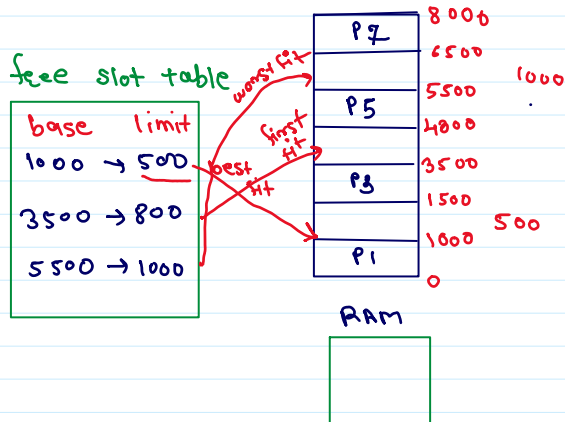
advantage - Simple to implement

Disadvantage - internal fragmentation

- fixed degree of mp.
- max size of process is also limited

② variable size partitioning

P8 1500



Advantages - no internal fragmentation
 .. NO fixed degree of mp
 - process size is limited

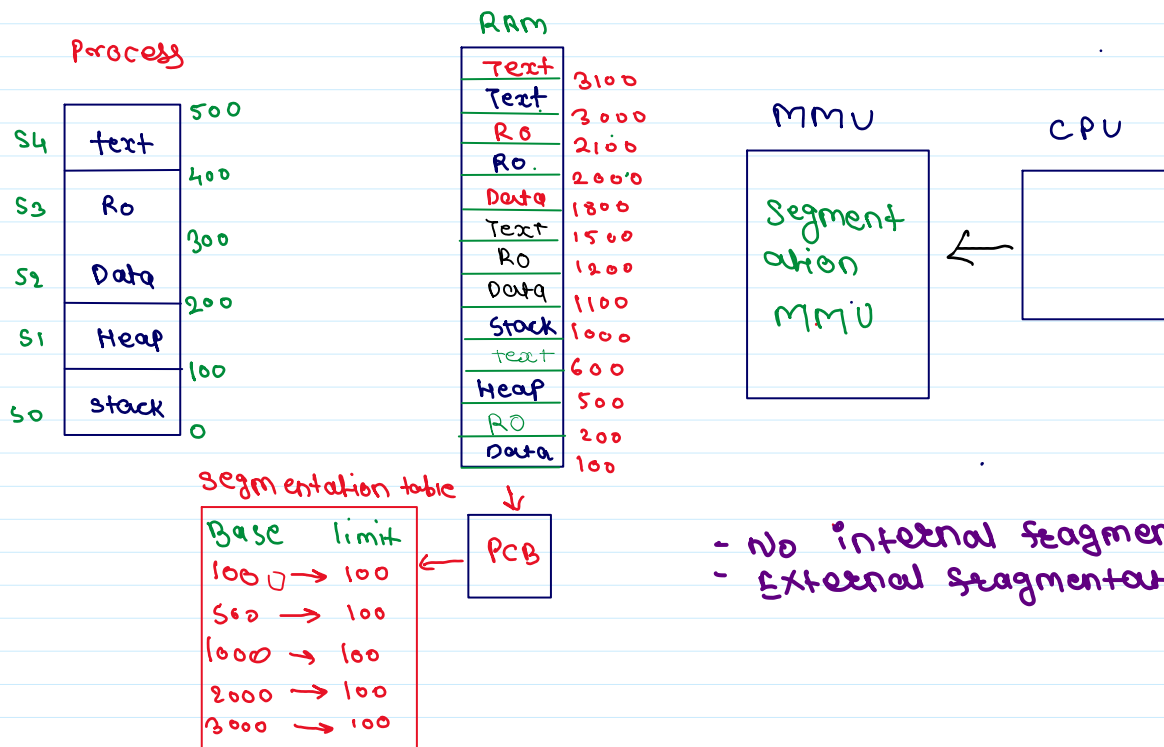
Disadvantages - External fragmentation

best fit -
 first fit -
 worst fit -

P9 200

P10 600

1) Segmentation



- No internal fragmentation
 - External fragmentation occurred.

* paging

RAM - frames.
Process - page.

4K

P1 (18Kb)

P5	2K
P4	2K
P3	4K
P2	4K
P1	4K

Page table

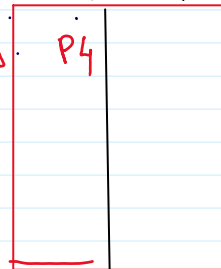
Page no	Frame
1	9v
2	11v
3	6v
4	3v
5	0v

RAM

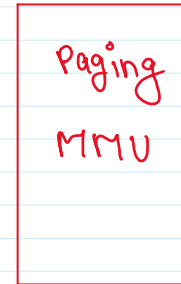
0	P5	0K
1	P1	4K
2	P2	8K
3	P4	12K
4	P3	16K
5	P4	20K
6	P3	24K
7	P1	28K
8	P2	32K
9	P1	36K
10	P3	40K
11	P2	44K
12	P1	48K
13	P2	52K
14	P3	56K
		60K

swap-out

HDD
virtual memory / swap area.



MMU



CPU.



* internal fragmentation occurred but min

* NO external fragmentation

CPU → P4 → Demand paging
- pure demand paging
- page fault